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PAPER

A general formulation of multi-dimensional space-charge-limited current density with nonzero monoenergetic initial velocity*

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Abstract

Thermionic energy converters (TECs) offer a direct method for converting heat into electricity, providing potential applications in waste heat recovery and renewable energy generation. While extensive research has focused on developing low work function materials, the performance of vacuum-based converters remains fundamentally hindered by space-charge effects, necessitating improved theoretical characterization of the space-charge limited current (SCLC). In this study, we derive exact theories for SCLC for multidimensional diodes with nonzero initial velocity (approximating high cathode temperatures) to demonstrate how to design TECs with higher steady-state currents to enhance the conversion of heat to electricity. We show that injected currents exceeding the SCLC limit result in a sharp decrease in the collector current, defining the bifurcation condition. Simulations using the two-dimensional particle-in-cell codes XOOPIC and VSim for both the SCLC and bifurcation regimes yield SCLC within 7% of the theoretical predictions. Finally, the sharp knee characteristic of the transition from one-dimensional thermionic emission to SCLC, known as the Miram curve, is extended to multidimensional diodes characterized by a single sharpness parameter σ, and its implications are discussed.

1. Introduction

Thermionic energy converters (TECs) convert heat directly to electricity without any moving parts, thereby eliminating losses due to mechanical work and achieving high efficiency of energy conversion [1]. These devices operate by forcing electrons through a gap by thermionic emission [2]. The electrons that are thermionically emitted by a hot emitter travel across a vacuum gap to a cold collector and an external load, where electrical work is done. Their high efficiency and low carbon emission make TECs potential candidates for clean energy [3].

The two main challenges affecting the performance of TECs are the limits on the current achievable due to space-charge buildup in the gap and the unavailability of electrodes with low work function that would enhance the release of electrons from the cathode into the gap [1]. Electron emission in a TEC relies heavily on the work function of the emitter's material. The individual work functions of the electrodes must be low, with a sufficiently large difference between those of the emitter and collector, for efficient TEC operation [4].

While several studies have characterized and developed materials with low work function [5–10], exact analytical models of space-charge effects that incorporate nonzero initial electron velocity have not been thoroughly developed for multidimensional TEC geometries. Beyond material optimization, active control strategies such as the application of longitudinal magnetic fields or the use of grid-controlled triode configurations have been proposed to actively suppress space-charge buildup and enhance current delivery [11–13]. While these active methods offer sophisticated pathways for maximizing device performance, our work focuses on deriving the fundamental analytical laws governing the passive electrostatic bottleneck in multidimensional diode geometries. Establishing this theoretical baseline is essential for characterizing the intrinsic transport limits that such advanced control technologies are designed to bypass. The space-charge limit hinders the functionality of TECs by limiting electron emission and reducing the current to the collector. As electrons are emitted, they are attracted to the positively charged emitter, surrounding it with a negative charge. This then causes subsequently emitted electrons to remain within this negative cloud, significantly altering the number of electrons arriving at the collector, which reduces the efficiency of the TEC. At a certain point, no more electrons can be injected into the gap, leading to the space-charge-limited current (SCLC), which represents the maximum current that can flow between the emitter and a collector in steady-state. In addition to optimizing TEC operation, better characterizing the SCLC is paramount for various applications, including electric propulsion, high-power microwave generation, microplasma formation, and various high-current devices [14, 15]. Mitigating space-charge effects often requires reducing the vacuum gap distance [16]; however, this presents practical difficulties because such small gaps are difficult to manufacture. Thus, characterization of SCLC in a multidimensional TEC is crucial for engineering better solutions.

Child [17] and Langmuir [18] derived the exact solution of the classical nonrelativistic SCLC density (SCLCD), with zero electron injection velocity, for a one-dimensional (1D) planar geometry over a century ago. The eponymous Child–Langmuir (CL) law for planar 1D SCLCD J_{CL} is given by [19]

$$J_{\mathrm{CL}} = \frac{4\epsilon_0}{9}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\frac{V_{\mathrm{g}}^{3/2}}{D^2}, \tag{1}$$

where ϵ_0 is the permittivity of vacuum, e and m_{e} are the charge and mass of the electron, respectively, and D is the separation distance between the grounded cathode and the anode held at a constant voltage V_{g}.

The CL law has been extended to account for nonplanar geometries [20–26], multiple dimensions [such as two-dimensional (2D) and three-dimensional (3D) planar diode geometries] [27–32], relativistic [33, 34], and quantum regimes [35–39]. Experiments showed that the current density does not follow the 3/2 dependence on voltage predicted by (1) when the electrons are emitted with nonzero injection velocities [40–43]. Jaffé derived an analytical solution to SCLCD in a 1D planar diode by modifying the CL law to account for monoenergetic emission [44]. This monoenergetic approximation is frequently treated as an effective characteristic velocity for the electrons that possess sufficient energy to cross the potential barrier. While the specific statistical definition of this proxy, such as the mean or root-mean-square velocity, can vary depending on the chosen mapping of the underlying thermal distribution, v_0 serves as the fundamental parameter governing the resultant kinetic energy of the electrons within the vacuum gap. If electrons are emitted from the cathode with an initial velocity v_0, the SCLCD in a 1D diode is given by [44]

$$J_{\mathrm{SCL,1D},v_0} = J_{\mathrm{CL}} f_{\mathrm{SCL}}(\beta), \tag{2}$$

where $\beta^2 = m_{\mathrm{e}}v_0^2/\left(2eV_{\mathrm{g}}\right)$ and $f_{\mathrm{SCL}}(\beta) = \left(\beta + \sqrt{1+\beta^2}\right)^3$. Further increasing the current beyond the SCLCD causes the virtual cathode to oscillate at roughly the beam plasma frequency, reflecting the electrons back to the emitter and sharply decreasing the current collected at the anode [45]. In a 1D planar diode, this threshold, called the bifurcation current density, is given in a 1D planar diode by [45]

$$J_{\mathrm{B,1D}} = J_{\mathrm{CL}} f_{\mathrm{B}}(\beta), \tag{3}$$

where $f_{\mathrm{B}}(\beta) = \left[\beta^{3/2} + \left(1+\beta^2\right)^{3/4}\right]^2$. The physical significance of the bifurcation point in our model is inherently linked to the material properties of the converter. In a self-biased TEC, the vacuum potential V_{g} is dictated by the difference between the emitter work function ϕ_{E} and the collector work function

ϕ_c (i.e. $eV_g = \phi_E - \phi_C$). Since the parameter β scales inversely with the square root of V_g, the transition between different potential profile regimes, or the bifurcation point, is physically determined by the choice of electrode materials and the resulting contact potential. Furthermore, the current enhancement discussed here does not require an external voltage; rather, the thermal energy of the electrons at the emitter surface provides the requisite initial velocity v_0 to partially mitigate space-charge effects, allowing the current to exceed the classical zero-velocity CL limit. Recently, we [46, 47] extended (2) and (3) to derive exact analytical solutions for the SCLCD and bifurcation current density for any 1D orthogonal diode using Lie symmetries [46] and variational calculus [47].

However, fundamentally characterizing and practically designing TECs requires a thorough understanding of SCLCD and bifurcation currents in multidimensional geometries (2D and 3D). In this paper, we derive exact analytical solutions for SCLCD and bifurcation current densities for various 2D and 3D planar geometries. We utilize the monoenergetic approximation, a standard approach in analytical beam physics [44], to resolve the cathode singularity of the CL law while maintaining analytical tractability that is lost with a full Maxwellian distribution. Specifically, we prove that the correction factors from (2) and (3) remain valid for *any* orthogonal diode geometry in *any* number of dimensions. In section 2, we mathematically derive SCLCD by relating the bifurcation potential to the vacuum potential, which we use to formulate an exact relationship between SCLCD and vacuum capacitance for any orthogonal geometry. We then derive analytical formulae for SCLCD and bifurcation current densities for various 2D and 3D geometries. In section 3, we present particle-in-cell (PIC) simulations using VSim and XOOPIC that agree with the theoretical predictions with a maximum error of approximately 7% for SCLCD and 7.6% for bifurcation current density. In section 4, we discuss the implications of the theory for TECs, considering the Maxwell–Boltzmann velocity distribution of the electrons due to finite temperature of the emitter. We make concluding remarks in section 5.

2. Theory

2.1. Relating SCLCD to vacuum capacitance

Consider a 1D planar diode geometry with a grounded cathode and an anode held at V_g, at a distance D from the cathode. The vacuum potential difference, denoted as V_g, is defined as the effective electrostatic potential drop across the vacuum gap between the cathode and the anode. In a practical TEC, this value is primarily determined by the difference in the work functions of the emitter and collector surfaces (i.e. $eV_g = \phi_E - \phi_C$), where ϕ_E and ϕ_C are work functions of the emitter and collector, respectively. Injecting electrons with a constant velocity v_0 causes the virtual cathode to form some distance from the cathode. We define SCLCD as the maximum steady-state current density transmissible across the diode gap for given V_g and v_0. Since the electric potential in the gap is a function of the distance from the cathode, the virtual cathode is defined as the location where the electric field normal to the cathode vanishes. The velocity of the electrons at the virtual cathode is in general not zero for space-charge limited (SCL) emission [45]. However, as the emitted current is increased beyond the SCLCD, the virtual cathode gains enough charge density to decelerate the emitted electrons to rest. The current density when the velocity of the electrons goes to zero at the virtual cathode is called the bifurcation current density and the corresponding potential (bifurcation potential) is represented by ϕ_B [45]. The bifurcation current density $J_{B,1D}$ for a D planar diode is given by [45]

$$J_{B,1D} = \epsilon_0 V_g^{3/2} \sqrt{\frac{2e}{m_e}} \frac{d^2\bar{\phi}_B}{dx^2} \sqrt{\bar{\phi}_B + \beta^2}, \tag{4}$$

where the dimensionless parameter $\bar{\phi}_B = \phi_B / V_g$ represents the normalized bifurcation potential in the gap. Moreover, the SCLCD is related to the bifurcation potential in the diode by [44]

$$J_{SCL,1D,v_0} = \epsilon_0 V_g^{3/2} \sqrt{\frac{2e}{m_e}} \frac{d^2\bar{\phi}_B}{dx^2} \sqrt{\bar{\phi}_B + \beta^2} \left(\frac{f_{SCL}(\beta)}{f_B(\beta)} \right). \tag{5}$$

Generalizing (4) to any multidimensional orthogonal geometry, we obtain the bifurcation current density j_B as

$$j_B = \epsilon_0 V_g^{3/2} \sqrt{\frac{2e}{m_e}} \nabla^2 \bar{\phi}_B \sqrt{\bar{\phi}_B + \beta^2}. \tag{6}$$

Generalizing (5) for any multidimensional orthogonal geometry yields

$$j_{\mathrm{SCL}} = \epsilon_0 V_{\mathrm{g}}^{3/2}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\nabla^2\bar{\phi}_{\mathrm{B}}\sqrt{\bar{\phi}_{\mathrm{B}}+\beta^2}\left(\frac{f_{\mathrm{SCL}}(\beta)}{f_{\mathrm{B}}(\beta)}\right). \tag{7}$$

Recently, Halpern *et al* maximized the bifurcation current in the gap by applying variational calculus and showed that for any orthogonal geometry, $\bar{\phi}_{\mathrm{B}}$ is given by [47]

$$\nabla^2\left(\bar{\phi}_{\mathrm{B}}+\beta^2\right) = \frac{\left|\nabla\left(\bar{\phi}_{\mathrm{B}}+\beta^2\right)\right|^2}{4\left(\bar{\phi}_{\mathrm{B}}+\beta^2\right)}. \tag{8}$$

It can be proved (see appendix for a proof) from (8) that $\bar{\phi}_{\mathrm{B}}$ is related to the normalized vacuum potential $\bar{\phi}_0$ in the gap by

$$\bar{\phi}_{\mathrm{B}}+\beta^2 = \left(-\beta^{3/2}+\sqrt{f_{\mathrm{B}}(\beta)}\bar{\phi}_0\right)^{4/3}. \tag{9}$$

Applying the Laplacian operator in general orthogonal coordinates to both sides of (9) and using $\nabla^2\bar{\phi}_0 = 0$ for vacuum potential (see appendix) gives

$$\nabla^2\bar{\phi}_{\mathrm{B}} = \frac{4f_{\mathrm{B}}(\beta)}{9}\left|\nabla\bar{\phi}_0\right|^2\left(-\beta^{3/2}+\sqrt{f_{\mathrm{B}}(\beta)}\bar{\phi}_0\right)^{-2/3}. \tag{10}$$

Substituting (10) and (9) in (6) yields

$$j_{\mathrm{B}} = \frac{4\epsilon_0}{9}V_{\mathrm{g}}^{3/2}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\left|\nabla\bar{\phi}_0\right|^2 f_{\mathrm{B}}(\beta). \tag{11}$$

Similarly, substituting (10) and (9) in (7) gives the SCLCD as a function of the magnitude of the normalized vacuum electric field $\left|\nabla\bar{\phi}_0\right|$ as

$$j_{\mathrm{SCL}} = \frac{4\epsilon_0}{9}V_{\mathrm{g}}^{3/2}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\left|\nabla\bar{\phi}_0\right|^2 f_{\mathrm{SCL}}(\beta). \tag{12}$$

Motivated by prior derivations of the CL law using charge-free capacitance [48], we take the volume average of (11) and (12) and noting that vacuum capacitance $C_0 = \epsilon_0\int\left|\nabla\bar{\phi}\right|^2\mathrm{d}V$ [28], gives the volume-averaged bifurcation current densities and SCLCD as

$$J_{\mathrm{B}} = \frac{\int j_{\mathrm{B}}\mathrm{d}V}{\int\mathrm{d}V} = \frac{4V_{\mathrm{g}}^{3/2}}{9}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\frac{C_0}{\int\mathrm{d}V}f_{\mathrm{B}}(\beta), \tag{13a}$$

and

$$J_{\mathrm{SCL}} = \frac{\int j_{\mathrm{SCL}}\mathrm{d}V}{\int\mathrm{d}V} = \frac{4V_{\mathrm{g}}^{3/2}}{9}\sqrt{\frac{2e}{m_{\mathrm{e}}}}\frac{C_0}{\int\mathrm{d}V}f_{\mathrm{SCL}}(\beta), \tag{13b}$$

respectively. Note that introducing capacitance provides a path toward deriving exact, analytic solutions for many geometries for which capacitance is known; however, it also leads to the SCLCD becoming volume-averaged. For planar geometries, the SCLCD and volume-averaged geometries are identical, but they may differ for more complicated geometries, making (11) and (12) more practically useful.

Substituting the volume-averaged SCLCD for zero electron injection velocity, defined by $J_{\mathrm{SCL}(0)} = 4V_{\mathrm{g}}^{3/2}C_0\sqrt{2e/m_{\mathrm{e}}}/(9\int\mathrm{d}V)$, in (13) gives

$$J_{\mathrm{B}} = J_{\mathrm{SCL},0}f_{\mathrm{B}}(\beta), \tag{14a}$$

and

$$J_{\mathrm{SCL}} = J_{\mathrm{SCL},0}f_{\mathrm{SCL}}(\beta), \tag{14b}$$

where $J_{\mathrm{SCL},0}$ corresponds to the volume-averaged SCLCD with $v_0 = 0$ for the geometry of interest. The mathematical factorization of SCLCD into a purely geometric component and a purely velocity-dependent component carries a distinct physical interpretation. The geometric configuration of the multidimensional diode governs the spatial topology of the vacuum electric field lines and determines the charge-free capacitance [28]. Conversely, injecting electrons with a monoenergetic initial velocity

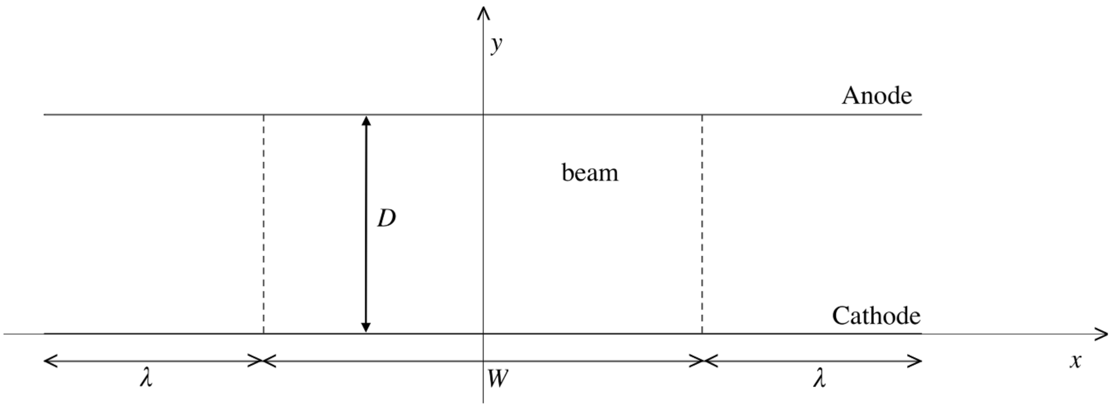

Figure 1. Geometry of the 2D planar diode with electron emission from the width W of the cathode located at $y = 0$ and centered at $x = 0$.

provides a uniform kinetic energy baseline across the entire emission surface. Because this initial kinetic energy uniformly elevates the potential barrier threshold required to form a virtual cathode, it scales the total charge capacity of the gap without altering the fundamental spatial paths governed by the electrode geometry. Consequently, the correction factors for initial velocity remain mathematically decoupled from the geometric form factor for any orthogonal diode configuration. Moreover, equation (14) can be used to obtain the bifurcation current density and SCLCD for *any* orthogonal geometry with monoenergetic emission, when the corresponding SCLCD for $v_0 = 0$ is known.

2.2. 2D planar geometries

We now consider a 2D planar geometry with the anode and cathode represented by the long strips at $y = D$ and $y = 0$, respectively, as shown in figure 1. Electron emission is restricted to a finite width W of infinite emissivity on the cathode. The lower surface of the cathode and the ledges of width 2λ do not emit electrons. An infinite magnetic field confines the motion of the electrons in the y -direction [27, 31].

To evaluate the specific physical case where electrons are injected with zero initial velocity ($v_0 = 0$), we set $\beta = 0$ in (8). This recovers the relationship derived by Darr and Garner [22] using the Euler-Lagrange equation as

$$\nabla^2 \bar{\phi} = \frac{|\nabla \bar{\phi}|^2}{4\bar{\phi}}, \tag{15}$$

where $\bar{\phi} = \phi / V_g$ is the normalized SCL potential. Denoting the normalized variables as $\bar{x} = x/D$ and $\bar{y} = y/D$, the boundary conditions for (15) are $\bar{\phi}\,(\bar{x}, \bar{y} = 0) = 0$ and $\bar{\phi}\,(\bar{x}, \bar{y} = 1) = 1$. The SCL potential obtained as the solution to (15) automatically satisfies the SCL boundary condition at the cathode since the normal component of the electric field at the cathode surface vanishes (i.e. $\mathrm{d}\bar{\phi}/\mathrm{d}\bar{y} = 0$ at $\bar{y} = 0^+$) even though that boundary condition is not required for the derivation [22]. In Cartesian coordinates, (15) can be written as

$$\frac{\partial^2 \bar{\phi}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{\phi}}{\partial \bar{y}^2} = \frac{1}{4\bar{\phi}} \left[\left(\frac{\partial \bar{\phi}}{\partial \bar{x}} \right)^2 + \left(\frac{\partial \bar{\phi}}{\partial \bar{y}} \right)^2 \right]. \tag{16}$$

Separating variables in (16) using $\bar{\phi}\,(\bar{x}, \bar{y}) = X(\bar{x})\, Y(\bar{y})$, we obtain the general solution to (16) as

$$\bar{\phi}\,(\bar{x}, \bar{y}) = a_0 (a_1 + 3\bar{y})^{4/3} + b_k \cosh^{4/3} \left\{ \frac{\sqrt{3}k}{2} (c_k + \bar{x}) \right\} \cos^{4/3} \left\{ \frac{\sqrt{3}k}{2} (d_k + \bar{y}) \right\}. \tag{17}$$

Applying the boundary conditions gives the general solution as a superposition of the solutions given in (17) as

$$\bar{\phi}\,(\bar{x}, \bar{y}) = \bar{y}^{4/3} + \sum_{n=1}^{\infty} c_n \cosh^{4/3} (\pi \bar{x}) \sin^{4/3} (\pi \bar{y}). \tag{18}$$

The constants c_n in (18) must be a function of $\bar{W} = W/D$ such that $\lim_{\bar{W}\to\infty} c_n \cosh^{4/3}(\pi \bar{W}/2) = 0$ to recover the electric potential for 1D (i.e. $\bar{\phi}_{1D} = \bar{y}^{4/3}$). Hence, we may write $c_n \cosh^{4/3}(\pi \bar{W}/2) \approx c_n \cosh(\pi \bar{W}/2) \ll 1$, when $\bar{W} \gg 1$. Thus, the infinite series in (18) converges in the limit of large $\bar{W}$. Assuming $\bar{W} \gg 1$, we may write the electric potential approximately as

$$\bar{\phi}(\bar{x},\bar{y}) \approx \bar{y}^{4/3}\left[1+\sum_{n=1}^{\infty} c_n \cosh(\pi \bar{x}) \frac{\sin^{4/3}(\pi \bar{y})}{\bar{y}^{4/3}}\right]. \tag{19}$$

Assuming the electric potential given by (19) and requiring the continuity of the electric field at the interface between the beam and vacuum region, Rokhlenko and Lebowitz [29] derived the approximate equation for the volume averaged SCLCD $J_{2D,SCL,0}$ at the cathode for $v_0 = 0$ when $W \gg D$ as $J_{2D,SCL,0}/J_{CL} = 1 + \left(0.19 + 0.48e^{-3.7\lambda/W}\right) D/W$, where 2λ is the total length of the ledges on the cathode that do not emit electrons. Substituting $J_{2D,SCL,0}/J_{CL}$ into (14*b*) gives the SCLCD with monoenergetic emission as

$$\frac{J_{2D,SCL,v_0}}{J_{CL}} = \left(1 + \frac{0.19 + 0.48e^{-3.7\lambda/W}}{W/D}\right) f_{SCL}(\beta). \tag{20}$$

Assuming $\lambda = 0$ and emission from the full front surface of the cathode, we recently derived the exact analytical solution for the volume averaged SCLCD as $J_{2D,SCL,v_0,\lambda=0} = F(\mu) D/[W F(m)]$ [28]. Using (14*b*), the volume averaged SCLCD for a monoenergetic injection velocity v_0 is

$$\frac{J_{2D,SCL,v_0,\lambda=0}}{J_{CL}} = \left[\frac{F(\mu)/F(m)}{W/D}\right] f_{SCL}(\beta), \tag{21}$$

where $F(\theta,x) = \int_0^\theta dt/\sqrt{1-x^2\sin^2 t}$ is the incomplete elliptic integral of the first kind with $F(x) = F(\pi/2,x)$. The modulus m and the corresponding complementary modulus $\mu = \sqrt{1-m^2}$ in (21) can be obtained for a given W/D using [28]

$$\frac{W}{D} = \frac{F(\mu)E(\omega,\mu) - E(\mu)F(\omega,\mu)}{[E(\mu)-F(\mu)]F(m) + F(\mu)E(m)}, \tag{22}$$

where $E(\theta,x) = \int_\theta^0 \sqrt{1-x^2\sin^2 t}\,dt$ is the incomplete elliptic integral of second kind, $E(x) = E(\pi/2,x)$, and the argument ω is given by

$$\sin^2\omega = \frac{F(\mu)-E(\mu)}{\mu^2 F(\mu)}. \tag{23}$$

It is also useful to obtain approximate closed-form solutions of (21) for large and small values of W/D. For $W \gg D$, we may approximate (21) to first order in W/D as

$$\frac{J_{2D,SCL,v_0,\lambda=0}}{J_{CL}} \approx \left[1 + \frac{1+\ln(2\pi W/D)}{\pi (W/D)}\right] f_{SCL}(\beta). \tag{24}$$

We recover the CL law (1) from (24) when $W/D \to \infty$ and $\beta \to 0$. For $W \ll D$, we may also approximate (21) as

$$\frac{J_{2D,SCL,v_0,\lambda=0}}{J_{CL}} \approx \left[\frac{\pi}{(W/D)\ln(4D/W)}\right] f_{SCL}(\beta). \tag{25}$$

2.3. 3D planar geometries

We next consider a 3D disk (3DD) geometry in which the anode and the cathode are represented by circular disks of radii R separated by a distance D. The anode is held at a constant voltage V_g, while the cathode is grounded. An infinite magnetic field restricts the beam to have a constant radius R. The SCLCD for zero injection velocity $J_{3DD,SCL,0}$ was derived recently using vacuum capacitance as [28]

$$\frac{J_{3DD,SCL,0}}{J_{CL}} = \frac{4D}{\pi R}\int_0^1 f(t)\,dt, \tag{26}$$

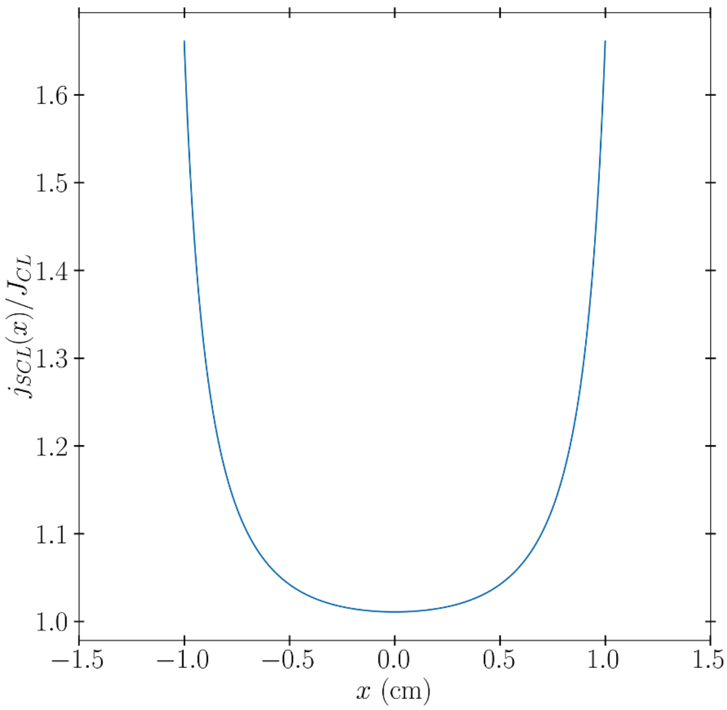

Figure 2. Current density as a function of location, $j_{\mathrm{SCL}}(x)$ scaled to the current density for the Child–Langmuir law, J_{CL}, along the cathode using (32) for initial velocity $v_0 = 0$. The sharp rises at the edges of the cathode ($x = \pm 1$ cm) correspond to the wings.

where $f(t)$ satisfies Love's integral equation, given by

$$f(t) - \frac{D}{\pi R}\int_{-1}^{1}\frac{f(\tau)}{\kappa^2 + (\tau - t)^2}\mathrm{d}\tau = 1. \tag{27}$$

Substituting (26) into (14b), we obtain the SCLCD for monoenergetic emission as

$$\frac{J_{\mathrm{3DD,SCL},v_0}}{J_{\mathrm{CL}}} = \left[\frac{4D}{\pi R}\int_{0}^{1} f(t)\,\mathrm{d}t\right] f_{\mathrm{SCL}}(\beta). \tag{28}$$

In addition to the exact SCLCD given by (28) for circular disks, we may write approximate solutions for $R \gg D$ to first order in R/D as

$$\frac{J_{\mathrm{3DD,SCL},v_0,R\gg D}}{J_{\mathrm{CL}}} \approx \left[1 + \frac{\ln(16\pi R/D) - 1}{\pi R/D}\right] f_{\mathrm{SCL}}(\beta). \tag{29}$$

Similarly, we may obtain SCLCD for $D \gg R$ by rewriting (28) as

$$\frac{J_{\mathrm{3DD,SCL},v_0,R\ll D}}{J_{\mathrm{CL}}} = \left[\frac{8}{\pi^2} + \frac{4D}{\pi R}\right] f_{\mathrm{SCL}}(\beta). \tag{30}$$

Similarly, for 3D plates with length and width equal to W, the volume-averaged SCLCD can be obtained from (14b) as [28]

$$\frac{J_{\mathrm{3D,\ plates,SCL},v_0}}{J_{\mathrm{CL}}} \approx \left[\frac{2F(\mu)/F(m)}{W/D} - 1\right] f_{\mathrm{SCL}}(\beta), \tag{31}$$

where the contribution of the fringing fields at the four corners to the SCLCD has been ignored and the incomplete elliptic integrals satisfy (22) and (23) for a given W/D [28].

3. Validation with PIC simulations

In this section, we assess the SCLCD as a function of location on the cathode and the development of bifurcation using PIC simulations using VSim and the theoretical predictions for the onset of bifurcation to PIC results using XOOPIC.

3.1. VSim simulations

To test the validity of the theory, we consider a specific case of a parallel plate geometry with a grounded cathode of length $L = 4$ cm, separated from an anode held at a constant voltage $V_g = 10$ kV by a gap distance $D = 1$ cm. An infinite strip of width $W = 2$ cm emits electrons with injection velocity v_0 from the cathode. The electron motion is restricted in the y -direction by a large magnetic field (100 T). While practical TECs necessitate microscale vacuum gaps to mitigate severe space-charge limitations, as detailed by Lo *et al* [49], we utilize these macroscopic dimensions in our VSim models strictly to computationally resolve and clearly demonstrate the bifurcation behavior. Note that Lo *et al* [49] benchmarked their planar 1D simulation to the Jaffé version of SCLCD for a monoenergetic initial velocity from (2) for their assessment of TECs. Because the governing equations are scale-invariant, confirming the theory at a macroscopic gap distance of 1 cm inherently validates its application to the microscopic dimensions required in functional operating devices. The SCLCD as a function of x along the cathode width can be written as an infinite series of hyperbolic cosines [see (19)] and is given by [29]

$$\begin{aligned}\frac{j_{\mathrm{SCL}}(x)}{J_{\mathrm{CL}}} &= 1 + 0.0106\cosh(3.881x) + 0.000264\cosh(6.675x)\\ &\quad + 8.577\times10^{-6}\cosh(10.065x) + 2.637\times10^{-7}\cosh(13.003x)\\ &\quad + 8.565\times10^{-9}\cosh(16.316x) + 2.748\times10^{-10}\cosh(19.306x)\\ &\quad + 7.645\times10^{-12}\cosh(22.582x) + 2.469\times10^{-13}\cosh(25.6x)\\ &\quad + 4.896\times10^{-15}\cosh(28.855x) + 1.589\times10^{-16}\cosh(31.891x)]\\ &\quad \times\left(\beta+\sqrt{1+\beta^2}\right)^3. \end{aligned} \tag{32}$$

The CL current density J_{CL} in (32) for this geometry is $J_{\mathrm{CL}} = 23\,339.5$ A m^{-2}. Figure 2 shows (32) for $\beta = 0$.

The 2D mean current density over the whole cathode for this geometry is given by

$$\frac{J_{\mathrm{SCL}}}{J_{\mathrm{CL}}} = \left[1 + \frac{\alpha}{W/D}\right]\left(\beta+\sqrt{1+\beta^2}\right)^3, \tag{33}$$

where $\alpha = 0.2067$ is obtained by taking the average of the current density given by (32) over the entire surface of the cathode [29]. Substituting $W = 2$ cm and $D = 1$ cm in (33), we obtain $J_{\mathrm{SCL}} = 25751.6\left(\beta+\sqrt{1+\beta^2}\right)^3$ A m^{-2}. We performed various runs in VSim by emitting some multiplier k times the current density given by (32), i.e. $j_e = k j_{\mathrm{SCL}}(x)$, such that $k = 1$ corresponds to the SCLCD. We varied k slowly from 0.5 to 1.5, and then back down to 0.5 for various values of β.

Figure 3 shows the anode and cathode currents as a function of k for various values of β. Just beyond $k = 1.0$, the current density at the anode drops suddenly to a bifurcation value that is noticeably lower than the peak value at $k = 1.0$. In this respect, the behavior in 2D is identical to that in 1D, with the transition slightly less abrupt than in 1D, although still rapid. The current in the 'bifurcation' regime is not steady-state, so the 'bifurcation' limit is a time-average over a quite noisy signal. Figure 4 shows the onset of the turbulent behavior of the current density in 2D.

Increasing β enhances the hysteresis and the transmitted current takes longer to return to the curve. Figure 5 shows the phase-space plot of y -component of the velocity of the electrons in the diode. For emitted current beyond the SCLC, the electrons are reflected from the virtual cathode, indicating the onset of bifurcation regime.

3.2. XOOPIC simulations

The bifurcation solutions are presented for various values of the dimensionless parameter β. In these comparisons, the case where $\beta = 0$ corresponds to the physical limit of zero initial electron injection velocity ($v_0 = 0$), providing a benchmark against the classical SCLCD. The 2D bifurcation solution is calculated with XOOPIC by measuring and averaging the current at the anode with an injection current $2-3\times$ higher than the theoretically predicted value. The numerical current data were recorded after the

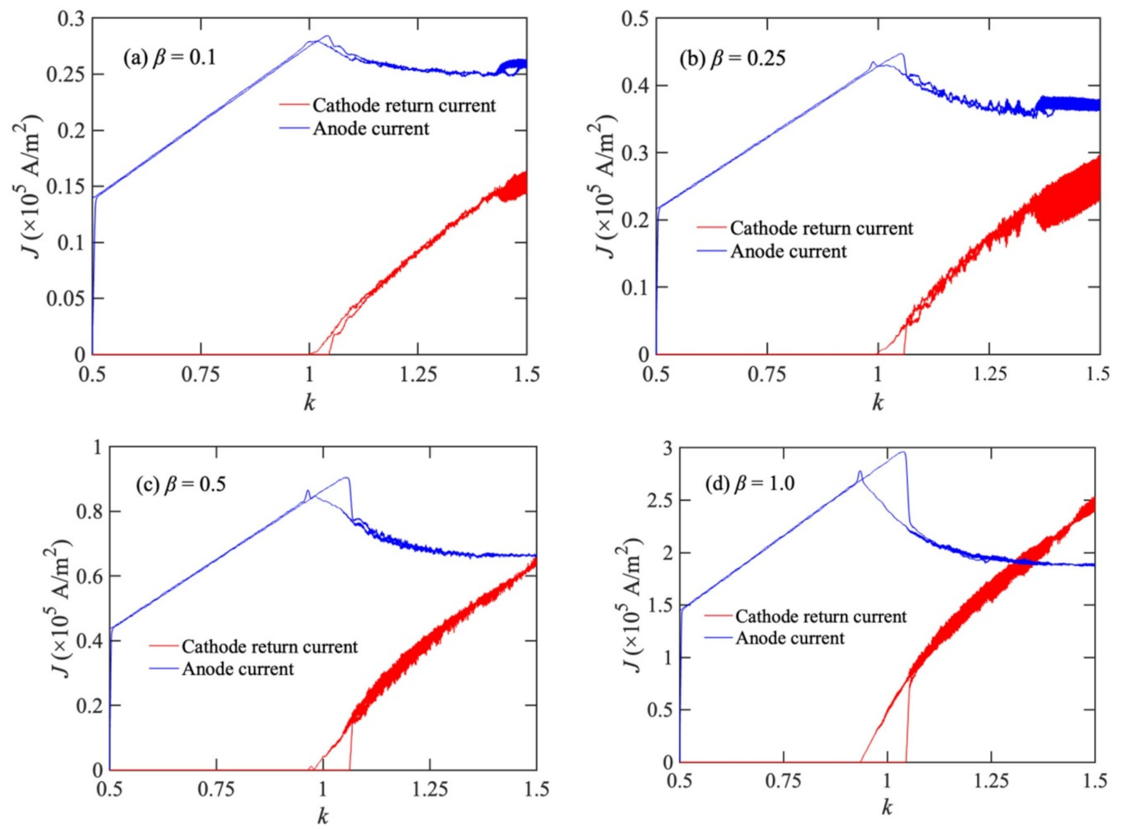

Figure 3. Anode and cathode current density J as a function of the multiplier k, which is defined such that $k = 1$ corresponds to the space-charge limit for (a) $\beta = 0.1$, (b) $\beta = 0.25$, (c) $\beta = 0.5$, and (d) $\beta = 1$. Bifurcation or virtual cathode oscillation occurs near $k = 1.0$, indicating that the maximum error with the theory is $\sim 7\%$.

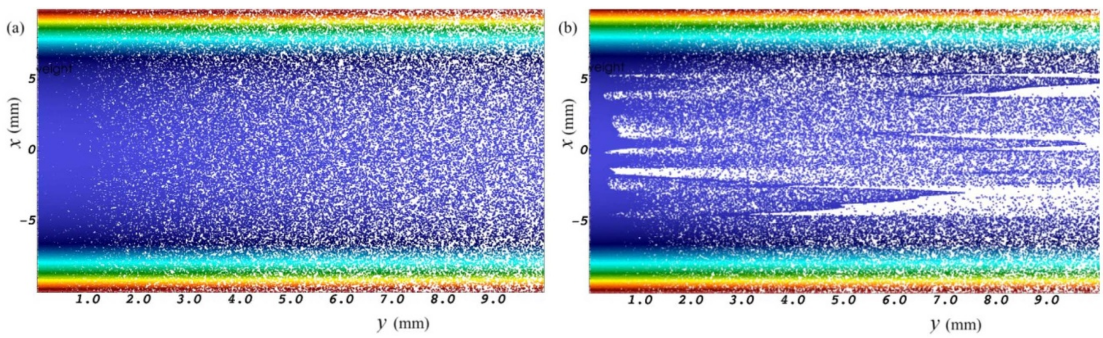

Figure 4. Particles in the laboratory space in VSim with the cathode on the left, anode on the right with (a) $k = 1.0$ (which corresponds to the space-charge limit) and (b) k slightly greater than 1.0 showing the non-steady flow in the 'bifurcation' regime. The color indicates the particle weight, so that the higher current density 'wings' are quite visible.

current stabilized. The initial velocity was initially set to 0.1 eV, which quickly damped the oscillations of the number and kinetic energy of the electrons [1]. More details on setting up XOOPIC are provided elsewhere [31].

Figure 6 compares the 2D bifurcation current density obtained from (13*a*) and XOOPIC for various β. Our simulations gave smaller relative differences when using smaller mesh sizes (and concomitant time step) for all W/D cases. The overall relative difference is below 10% and might be further reduced by using finer mesh cells, especially for small W/D.

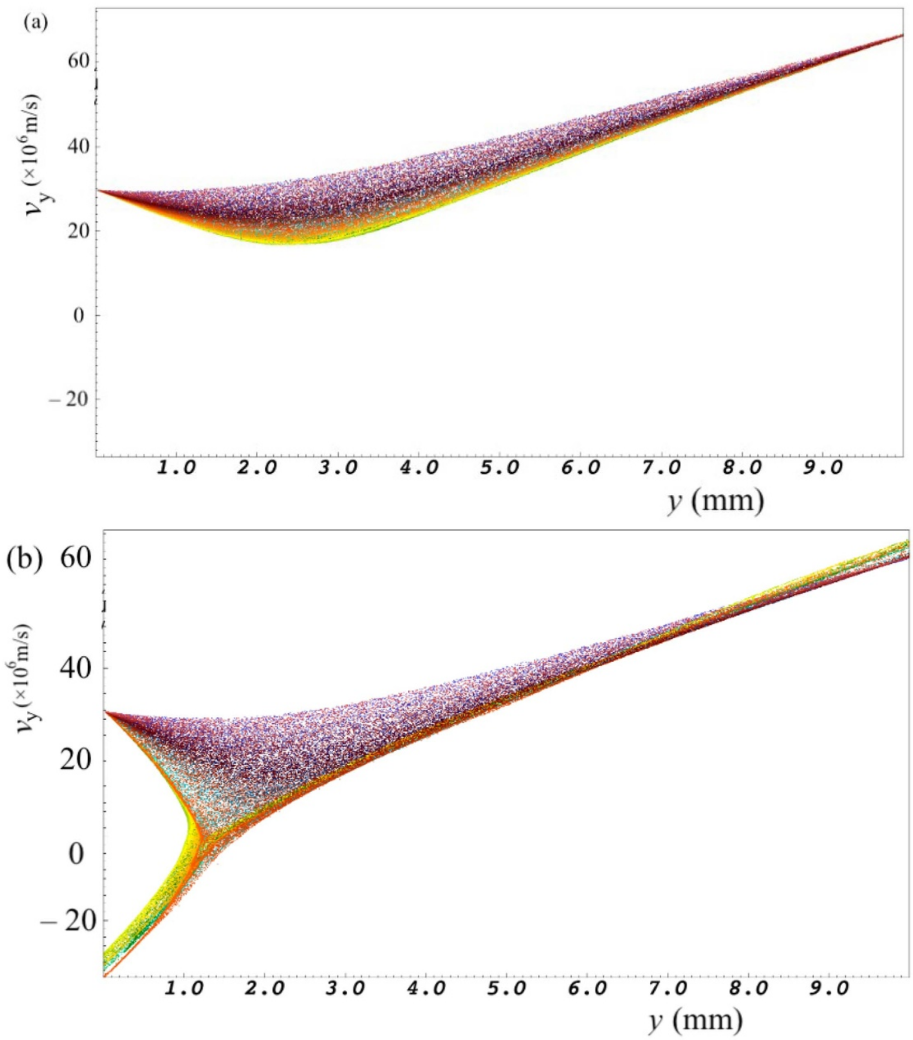

Figure 5. Particles in the laboratory space with the cathode on the left and the anode on the right with (a) $k = 1.0$ and (b) k slightly greater than unity showing reflection of electrons from the virtual cathode above the space-charge limit. The color indicates the particle weight, so that the higher current density 'wings' are quite visible.

4. Implication for TECs

Miram curves are important and widely used to characterize cathode performance of an electron emitting source, such as a TEC. To contextualize these metrics within practical device engineering, high-performance TECs typically operate with emitter temperatures ranging from 800 °C to 1200 °C, yielding current densities between 1 and 8 A cm^{-2} and necessitating vacuum gaps on the order of 1–100 μm to mitigate severe space-charge limitations. These $J - T$ curves plot the cathode averaged current density as a function of the cathode temperature of the TEC [50]. In 1D, J transitions sharply from thermionic emission (i.e. the Richardson–Laue–Dushman equation [51]) to the SCLC (i.e. CL in vacuum) at a critical temperature T_c. While many studies have examined the Miram curves for 1D, a theoretical derivation of the cathode averaged current density has not been developed for 2D and 3D cathodes [52]. The monoenergetic approximation utilized in this study provides exact analytical solutions for these multidimensional geometries by acting as a proxy for the average thermal velocity of electrons overcoming the virtual cathode. We acknowledge inherent limitations in this approach; the model does not account for the full Maxwell–Boltzmann velocity distribution, the complex patchiness of work functions on polycrystalline emitter surfaces, or potential collisional effects in non-ideal vacuums. Establishing these boundary conditions allows us to mathematically formulate the transition.

It is important to emphasize that the theoretical formulations for the multidimensional SCLC derived herein govern the macroscopic vacuum transport of electrons. Consequently, these specific analytical solutions are fundamentally independent of the electrode material composition, relying solely on the device geometry, applied voltage, and initial injection velocity. However, the practical manifestation of

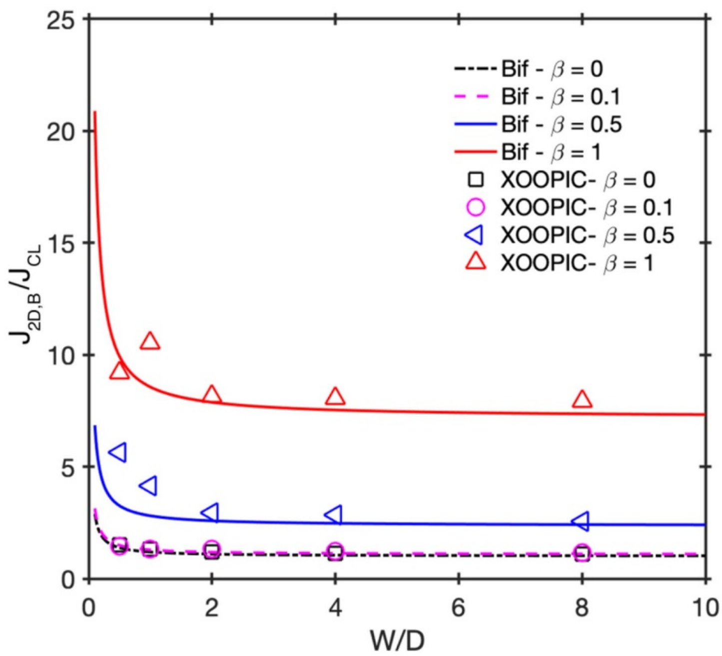

Figure 6. Comparison between 2D bifurcation current density, $J_{2D,B}$, obtained using XOOPIC and theory for various values of β, normalized to the 1D CL current density, J_{CL}. For these simulations, $D = 1$ cm, $V_g = 10$ kV, and $J_{CL} = 23\,339.5$ A m^{-2}.

the transition knee in a physical device relies heavily on the material-dependent thermionic emission current dictated by the chosen electrode. Common physical examples of these geometrical configurations include large-area parallel plates utilizing standard barium-dispenser tungsten or lanthanum hexaboride cathodes, which serve as excellent approximations for 1D planar geometries. For multidimensional applications, 2D planar geometries effectively represent finite-width ribbon or strip emitters, while 3D planar geometries correspond to finite circular disks or patterned micro-emitter arrays leveraging the advanced low-work-function materials discussed previously.

The sharp knee of a 1D TEC can be written as

$$J_{1D} = \frac{J_T + J_{SCL} - |J_T - J_{SCL}|}{2}, \tag{34}$$

where J_T is given by Richardson–Dushman equation $J_T = AT^2 \exp\left[-\phi/(kT)\right]$. Equation (34) gives a sharp knee since $J_{1D} = J_{SCL}$ for $J_T > J_{SCL}$ and $J_{1D} = J_T$ for $J_T < J_{SCL}$. In a strictly 1D planar diode, the entire cathode surface transitions from temperature-limited emission to space-charge-limited emission at the exact same critical temperature. This simultaneous global transition mathematically corresponds to taking the exact minimum of the two current regimes, which can be expressed using an absolute value function as shown in (34). However, in two-dimensional and 3D geometries, fringing electric fields and finite edge effects cause the local emission current density to vary across the cathode surface. Consequently, as the temperature increases, the central regions of the cathode may reach the space-charge limit while the outer edges remain temperature-limited. This spatial nonuniformity smears the global transition of the cathode-averaged current. To capture this smooth transition mathematically, we replace the sharp absolute value function with a hyperbolic smoothing function. This introduces a single sharpness parameter σ characterizing the smooth transition, which yields

$$J_{2D} = \frac{J_T + J_{SCL} - \sqrt{(J_T - J_{SCL})^2 + \sigma^2}}{2}, \tag{35}$$

where σ is the sharpness parameter. We note that $\sigma \to 0$ as $W/D \to \infty$ to recover the sharp knee of 1D cathodes. Hence, σ is a function of W/D, i.e. $\sigma \equiv \sigma(W/D)$. Substituting (14*b*) in (35), we obtain the normalized current density as

$$\frac{J_{2D}}{J_{SCL,0}} = \frac{\frac{J_T}{J_{SCL,0}} + f_{SCL}(\beta) - \sqrt{\left(\frac{J_T}{J_{SCL,0}} - f_{SCL}(\beta)\right)^2 + \frac{\sigma^2}{J_{SCL,0}^2}}}{2}. \tag{36}$$

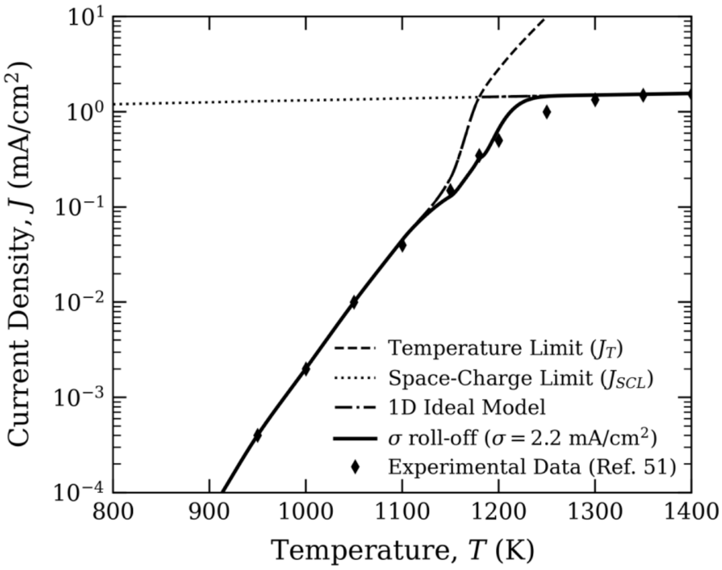

Figure 7. Current density J as a function of cathode temperature T. The dashed curve denotes the temperature-limited current J_{T}, the dotted curve denotes the space-charge-limited current J_{SCL}, and the dash-dotted curve shows the ideal 1D sharp-knee model $J = \min(J_{\mathrm{T}}, J_{\mathrm{SCL}})$. Filled diamond symbols represent the experimental data from Reference [52]. The solid curve shows the smooth roll-off using $\sigma = 2.2$ mA cm^{-2}.

Since $J_{\mathrm{SCL},0}$ and $f_{\mathrm{SCL}}(\beta)$ are known for a given device configuration and temperature (or v_0), we can determine the sharpness parameter σ from the experimental data of Miram curves. To illustrate the utility of this model, we present two examples from the literature.

First, we examine foundational data from the 1982 diagnostic study by Cattelino, Miram, and Ayers, which generated performance curves for a classic macroscopic Type MM dispenser cathode [50]. For a designated fully SCLCD of 4 A cm^{-2}, the ideal 1D model predicts a sharp transition knee at a specific critical temperature. However, the measured device exhibits a smooth roll-off due to non-ideal emission characteristics. An experimentalist can evaluate the multi-dimensional geometry of the device by extracting the sharpness parameter directly from this empirical Miram curve. At the theoretical transition temperature where the Richardson–Dushman temperature-limited current equals the theoretical vacuum space-charge limit, the measured multi-dimensional current density will be lower than the ideal limit. Rearranging (35) for the condition where $J_T = J_{SCL}$ yields $\sigma = 2(J_{SCL} - J_{2D})$. For example, for $J_{\mathrm{SCL}} = 4$ A cm^{-2}, we can estimate the measured current J_{2D} directly from the graphical data in figure 1 of their study [50]. The plot indicates a current of approximately 3.2 A cm^{-2} at that exact transition temperature, so the sharpness parameter σ evaluates to 1.6 A cm^{-2}.

To further demonstrate the physical relevance of this model for modern microscale devices, we apply the same procedure to experimental data from a microfabricated TEC developed by Lee *et al* [51] Their study characterizes a barium-coated silicon carbide emitter with an area of 500 μm $\times$ 500 μm. As illustrated in figure 7, the measured emission saturates at an absolute space-charge limit of approximately 5 μA, yielding a theoretical multidimensional J_{SCL} of approximately 2 mA cm^{-2}. For this specific microscale architecture, the 1D ideal model predicts a sharp transition knee at roughly 1400 K. However, at this theoretical transition temperature, the measured current density rolls off to approximately 0.4 mA cm^{-2} due to localized multidimensional space-charge constraints. Applying the same algebraic rearrangement of (35) yields a sharpness parameter σ of 2.2 mA cm^{-2}. Using this single extracted parameter, the model captures the smooth transition region in which space-charge effects govern the emission roll-off. This supports the scale-invariant character of the formulation and suggests that it can describe multidimensional transport in both macroscopic vacuum devices and microfabricated architectures.

5. Conclusion

TECs offer a promising avenue for direct conversion of heat into electricity, with potential applications in various fields, such as space exploration, waste heat recovery, and renewable energy generation. In addition to its importance in optimizing TECs, characterizing SCLCD with monoenergetic emission is crucial for characterizing many high-power and high-current devices. We have presented exact and closed-form approximations to SCLCD for various 2D and 3D diode geometries. The simulation results agree well with theoretical predictions. The equations developed to characterize the sharp knee of 1D thermionic cathode were extended to multiple dimensions with a single sharpness parameter that can be determined experimentally.

While this study assumed monoenergetic emission, which is consistent with assumptions used to benchmark PIC simulation studies of TECs [49, 52], future work could extend this to include a Maxwellian distribution of the velocity for electrons emitted from a hot cathode. Furthermore, these theoretical results can be used to guide a new PIC algorithm to numerically obtain the cathode averaged current densities in various diode geometries. Ultimately, combining the SCLCD formulation derived here with nexus theories that bridge space-charge effects and emission equations (e.g. Fowler–Nordheim and Richardson–Laue Dushman) will offer critical insights for multidimensional device design [53]. Moreover, the release of contaminants or leakage of vacuum components could result in imperfect vacuum, necessitating the incorporation of collisions into the transition between electron emission mechanisms and SCLCD [54] and the application of SCLCD generalized for collisions rather than the vacuum SCLC given by J_{CL} [55]. Also, for smaller TECs at non-vacuum pressures, the additional electrons emitted may reduce the threshold for breakdown, leading to another potential avenue for device failure that may require additional study [56, 57]. Additionally, the exact analytical solutions for multidimensional geometries derived here are directly applicable to the design of high-performance TECs utilizing micro/nanoscale spacers. As reviewed by Loubet *et al* [58], minimizing the vacuum gap is essential for mitigating space-charge effects, but incorporating dielectric spacers to maintain these gaps introduces complex 3D geometrical constraints that deviate from ideal 1D behavior. Our formulation allows for the precise calculation of SCLC in such spacer-filled gaps, facilitating the optimization of spacer shape and distribution to maximize electron transport while balancing trade-offs with parasitic thermal conduction.

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Data availability statement

All data that supports the finding of this study are included within the article.

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Appendix

We consider an orthogonal diode geometry with the generalized metric $ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$, where the coordinates are represented by q_i and the corresponding scaling factors by h_i. The normalized vacuum potential $\bar{\phi}_0$ is defined such that $\nabla^2 \bar{\phi}_0 = 0$ for a given orthogonal geometry. The Laplacian of a linear function of vacuum potential $g(\bar{\phi}_0) = (a + b\bar{\phi}_0)^{4/3}$ in orthogonal coordinates is given by

$$\nabla^2 g = \frac{1}{h_i h_j h_k} \sum_{(i,j,k)} \frac{\partial}{\partial q_i} \frac{h_j h_k}{h_i} \left(\frac{\partial (a + b\bar{\phi}_0)^{4/3}}{\partial q_i} \right), \tag{A1}$$

where the summation is taken cyclically over coordinates in the metric, i.e. $(i,j,k) \rightarrow (j,k,i) \rightarrow (k,i,j)$, and a and b are independent of q_i. Expanding (A1) yields.

$$\nabla^2 g = \frac{1}{h_i h_j h_k} \sum_{(i,j,k)} \frac{\partial}{\partial q_i} \frac{h_j h_k}{h_i} \left(\frac{4b}{3} g^{1/4} \frac{\partial \bar{\phi}_0}{\partial q_i} \right). \tag{A2}$$

Simplifying (A2), we obtain

$$\nabla^2 g = \frac{4b}{3} g^{1/4} \nabla^2 \bar{\phi}_0 + \frac{4b^2}{9} \left| \nabla \bar{\phi}_0 \right|^2 (a + b\bar{\phi}_0)^{-2/3}. \tag{A3}$$

Applying $\nabla^2 \bar{\phi}_0 = 0$ to (A3) gives

$$\nabla^2 g = \frac{4b^2}{9} \left| \nabla \bar{\phi}_0 \right|^2 (a + b\bar{\phi}_0)^{-2/3}. \tag{A4}$$

Next, the square of the magnitude of the gradient of g in orthogonal coordinates is given by

$$|\nabla g|^2 = \sum_{(i,j,k)} \frac{1}{h_i^2} \left(\frac{\partial (a + b\bar{\phi}_0)^{4/3}}{\partial q_i} \right)^2. \tag{A5}$$

Carrying out the partial derivative in (A5) and simplifying yields

$$|\nabla g|^2 = \frac{16b^2}{9} \sum_{(i,j,k)} \frac{(a + b\bar{\phi}_0)^{2/3}}{h_i^2} \left(\frac{\partial \bar{\phi}_0}{\partial q_i} \right)^2. \tag{A6}$$

Simplifying (A6) by performing the sum over the three coordinates, we obtain

$$|\nabla g|^2 = \frac{16b^2}{9} (a + b\bar{\phi}_0)^{2/3} |\nabla \bar{\phi}_0|^2. \tag{A7}$$

Solving for $\left| \nabla \bar{\phi}_0 \right|^2$ in (A7) and substituting into (A4) gives

$$\nabla^2 g = \frac{|\nabla g|^2}{4g}. \tag{A8}$$

Hence, for any function $g=\left(a+b\bar{\phi}_0\right)^{4/3}$ of normalized vacuum potential $\bar{\phi}_0$, we may rewrite (A8) as

$$\nabla^2\left(a+b\bar{\phi}_0\right)^{4/3}=\frac{\left|\nabla\left(a+b\bar{\phi}_0\right)^{4/3}\right|^2}{4\left(a+b\bar{\phi}_0\right)^{4/3}} \tag{A9}$$

Comparing (A9) and (8), we may write, without loss of any generality,

$$\bar{\phi}_{\mathrm{B}}+\beta^2=\left(a+b\bar{\phi}_0\right)^{4/3}. \tag{A10}$$

The constants a and b in (A10) can be determined by the boundary conditions at the cathode (represented by $\mathcal{C}$) and the anode (represented by $\mathcal{A}$) in general orthogonal coordinates by

$$\bar{\phi}_0(\mathcal{C})=0;\ \bar{\phi}_0(\mathcal{A})=1 \tag{A11}$$

for the space-charge limit and

$$\bar{\phi}_{\mathrm{B}}(\mathcal{C})=0; \bar{\phi}_{\mathrm{B}}(\mathcal{A})=1 \tag{A12}$$

for the bifurcation condition. Evaluating (A10) at $\mathcal{C}$ using (A11) and (A12) gives $a=-\beta^{3/2}$; evaluating (A10) at $\mathcal{A}$ and using (A11) and (A12) gives $b=\beta^{3/2}+\left(1+\beta^2\right)^{3/4}=\sqrt{f_{\mathrm{B}}(\beta)}$. Substituting a and b in (A10) yields

$$\bar{\phi}_{\mathrm{B}}+\beta^2=\left(-\beta^{3/2}+\sqrt{f_{\mathrm{B}}(\beta)}\bar{\phi}_0\right)^{4/3}. \tag{A13}$$

This completes the proof of the functional relationship between $\bar{\phi}_{\mathrm{B}}$ and $\bar{\phi}_0$, given by (9) in the main text.

Next, taking the Laplacian of (A13) in orthogonal coordinates, noting that $\nabla^2\beta=0$, and using (A4), we obtain

$$\nabla^2\bar{\phi}_{\mathrm{B}}=\frac{4f_{\mathrm{B}}(\beta)\left|\nabla\bar{\phi}_0\right|^2}{9}\left(-\beta^{3/2}+\sqrt{f_{\mathrm{B}}(\beta)}\bar{\phi}_0\right)^{-2/3}. \tag{A14}$$

This completes the proof of (10) in the main text.

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